

CFPB Consumer Complaints - Predictive Modeling and Text Analysis Using Supervised and Unsupervised Learning

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1. Introduction

The Consumer Complaints Database provided by the Consumer Financial Protection Bureau (CFPB) compiles consumer complaints related to financial services and financial products. These complaints are passed along to their respective organization, where the financial institutions respond to the complaint, typically with one of a few options: Closed, Closed with Explanation, Closed with Non-Monetary Relief, Closed with Monetary Relief. The problem we want to address is how we can provide consumers with a better understanding of the possibility that their complaint will result in some form of financial restitution.

Some of us have worked on customer-focused initiatives at large banking institutions in the past. In the spirit of helping customers, we want to develop the data science foundations of a tool that could be used to better inform consumers about their complaints as they relate to receiving monetary compensation for their issue or just a general response from the financial institution. Given this context, our goal is to develop our model in such a way that it both provides consumers with good quality predictions for whether they will receive compensation and simultaneously does not discourage them from submitting valid consumer complaints regarding any other difficulties they are facing.

To build out this tool, we developed a two-stage approach in our supervised modeling. The first model run produces results on a base set of features. Then, using features generated from unsupervised modeling, we inform a final supervised model to see the difference that consumer narratives have on our target prediction task. For supervised modeling, we compare four different models: Logistic Regression, KNN, Random Forest, and SVM, evaluating them on their F1 scores to choose the best model. In addition to generating features for the prediction task, we use unsupervised methods to perform data mining on the text of the consumer narratives to understand themes across the complaints. We utilize K-Means clustering and Latent Dirichlet Allocation (LDA) as they are widely used for text modeling. While several of the works described in **Section 2** also explore topics in the complaint database using LDA, our project is unique in its use of pyLDAviz to surface rarer terms within topics.

In the initial supervised modeling task that builds on base features and sentiment, the Random Forest model produced the best results. From this model, we found that the primary signal exists among categorical features, including the Company, Issue, Product, and Sub-Issue. Although the F1 score result was only 0.45, after evaluating the model's performance, we found that there was a rather consistent tradeoff between precision and recall. From our evaluation, we concluded that the quality of features in terms of the signal they are able to generate was the main limiting factor of model performance.

Applying unsupervised learning techniques to the text of the complaint narratives revealed some interesting themes, despite signaling homogeneity in the topic space overall. Our best LDA model performed moderately well (coherence score: 0.51), while our best K-Means model did not produce strong cluster separation (silhouette score: 0.024). Out of the 20 topics generated by our LDA model, 50% of them were informative, while the rest were characterized by product names like "check" and "credit card." The informative topics gave more specific details about the subjects of the complaints, and by applying sampling and visualization methods, we were able to uncover less obvious themes.

2. Related work

Related Work 1: *Latent Dirichlet allocation (LDA) for topic modeling of the CFPB consumer complaints* (Bastani, et al., 2019)

- This paper applies LDA to the CFPB Consumer Complaints Database to understand themes across the narratives and plot changes over time.
- Our project also applies LDA, but has a larger scope in that it also utilizes supervised learning to predict monetary relief.

Related Work 2: *Predictive Analysis of CFPB Consumer Complaints Using Machine Learning* (Vaishnav et al., 2024)

- Vaishnav et al. (2024) analyzes the CFPB database on two parallel tracks: LDA to identify themes and supervised learning to predict outcomes.
- Our supervised learning models focus on binary classification by creating two outcome buckets: monetary and non-monetary relief, while their model performs multi-class prediction using the eight possible outcome values in the database.

Related Work 3: *Predictive model for CFPB consumer complaints* (Nalluri, V., 2023)

- Nalluri (2023) trains several models, including Random Forest, to predict outcomes for mortgage-related complaints in the CFPB database made against Wells Fargo.
- Our project includes all companies in the database and has a broader product scope than only mortgages. Further, we apply unsupervised learning methods, while Nalluri does not.

3. Data Source

The Consumer Complaints Database is accessible from the [CFPB's website](#). The database provides all information in a single transactional database that gets updated over the course of the complaints lifecycle. Because of this, the data contains both closed historical complaints and current in-progress complaints. For retrieval, they offer options to download the data via CSV or JSON, along with an API to access certain endpoints. For our project, we exclusively used the CSV download to develop our application. Our final download was performed in June 2026, and we filtered our dataset to the last decade ending with `May 31, 2026`.

While the total number of records in the database is well over 10 million, we decided to focus specifically on consumer banking-related products that we have encountered in our careers. Specifically, we decided to focus on the categories of "Mortgage", "Checking or savings account", "Credit card", "Credit card or prepaid card", and "Bank account or service". After filtering to this subset of banking products, we ended up with roughly 1.1 million complaint records with 19 columns.

Among the available columns, we chose to focus on the columns of "Product", "Sub-product", "Issue", "Sub-issue", "Company", "Submitted via", "Date received", and "Consumer complaint narrative". Additionally, due to the mixture of historical and in-progress records and because fields are updated at different times, we had to carefully evaluate columns for potential data leakage. To avoid any potential leakage in our models, we excluded the fields "Timely response", "Date sent to company", "Company response to consumer", "Company public response", "Consumer disputed?", and "Consumer consent provided?" as all these columns appeared problematic for our prediction task or simply do not exist at the time of prediction given that our prediction task is oriented towards when the consumers initially submit their complaints rather than later on in the complaints lifecycle.

4. Feature Engineering

For supervised model development, the feature engineering process focused on turning the raw CFPB complaint fields into usable predictors that enabled interpretability and were also computationally efficient. To begin, we used the complaint attributes that were used for the categorical assignment of the complaint record. This includes the fields mentioned above for "Product", "Sub-product", "Issue", "Sub-issue", "Company", and "Submitted via". In addition to those categorical features, we derived date-related features for things like year, month, or day of the week. Finally, for a simple text baseline, we intentionally avoided significant narrative feature generation inside the initial model development and only created some lightweight features in the training notebook or pre-computed sentiment. In the end, for our base model, we created a boolean indicator for whether a narrative was present, a measure for narrative length, and a precomputed VADER sentiment score.

Among all our categorical features, null values were a reality we had to face. For our baseline model, we wanted to keep these in to provide insight into how they degrade model results, if at all, relative to our

later supervised run that only used fully populated features. Therefore, in our base model, we filled in missing values with the categorical value of “missing” rather than imputing a common categorical feature, since we wanted to avoid overfitting on any specific category.

Additionally, to keep our feature space from becoming too large, especially for some of our columns with high cardinality, the rare categories that were uncommon were grouped by capping the number of categorical encodings that could be made. Because of this, our model seeks to generalize by not overfitting on low-volume categories that may not have enough data to provide a meaningful prediction.

After all these feature engineering steps, the final preprocessing pipeline was then able to handle all our numeric and categorical columns encoding so that the same feature engineering logic got consistently applied across our four candidate models.

For unsupervised learning, we dropped blank consumer narratives as they would have no text data to analyze. We also dropped rows with null values in the categorical columns that would be available at the time of prediction, such as Product, Sub-product, Issue, and Sub-issue. After sampling short narratives, we decided to filter out entries less than 30 characters (phrases like “See the attached documents”) as they would not add much context for text modeling. We cleaned and pre-processed the remaining narratives by normalizing whitespace, lowercasing, lemmatizing, and tokenizing the text. No encoding issues were found. After removing 21,631 duplicate narratives, we vectorized the processed text on two parallel paths, applying TF-IDF for K-Means, and CountVectorizer for LDA. In total, 398,004 narratives were used as features.

5. Supervised Learning

5.1 Methods Description

Our supervised learning workflow focuses on predicting whether a CFPB consumer banking complaint resulted in monetary relief from the company. Using the final processed relief dataset, we ran a model using existing complaint categories plus some lightweight narrative indicator variables. Then, we also ran a model with generated features from our unsupervised LDA topic analysis to see if more advanced narrative features would improve performance. The target variable is the same for both models, with complaint labels as “Closed with monetary relief” as the positive class, and all other complaint outcomes as the negative class (no monetary relief). The model we review in the following section is the baseline model without unsupervised features. For performance on our alternative model exploration with unsupervised model feature results, see Appendix B.

For model training, we began with a stratified train/test split to keep our observation proportions consistent, as we had to downsample our dataset in order to improve the training efficiency during computation. Then, to train, we used cross-validation within the training data to compare models and tune hyperparameters across a grid search. This separation allowed us to get a cross-validated model to select our final performative model, in addition to keeping a final holdout set for downstream evaluation in both the development and evaluation notebooks.

We explored four different model families in order to compare methods with different learning mechanisms. We chose to do all four as we wanted to have a solid basis for comparing each model family type. First, we used Logistic Regression as our linear probabilistic baseline model because it is fast, interpretable, and often performs well and is timely on higher-dimensional data like our encoded categorical data. Second, we used K-Nearest Neighbors (KNN) as an instance-based method to test whether local similarity among complaint observations could support prediction. However, due to performance issues with both KNN and Random Forests, our final run limited the KNN complexity as it routinely performed the worst in both timeliness and performance. Third, we used a Support Vector Machine (SVM) to evaluate as a non-probabilistic decision margin method, as it can sometimes separate classes more effectively than simpler linear models. Lastly, we used a Random Forest classifier as our tree-based ensemble because it can model complex relationships between complaint characteristics, company information, and complaint issues. Unlike linear models, Random Forests can automatically

capture interactions among variables and often perform well when the underlying relationships are nonlinear. For these reasons, we were not surprised that it performed the best out of all four models.

Feature representations focused on the structured features provided in each complaint. These features include: “Product”, “Sub-product”, “Issue”, “Sub-issue”, and “Company”. Additionally, we included lightweight narrative length features and a precomputed VADER sentiment polarity score. We intentionally avoided heavy feature generation in our training notebook for the final supervised models in order to keep the workflow computationally manageable and clean. One area of significant adjustment was that to avoid excessive sparsity in high cardinality categorical features, rare levels were grouped together by applying category limits before final encoding. This lets our models focus on the most common situations and improves model training times.

With our models, each went through hyperparameter tuning performed via cross-validation grid search for each model family. For Logistic Regression, we explored regularization strength. For KNN, we varied neighborhood size and distance settings. For SVM, we tested a small set of regularization values and kernel settings to see performance improvements without creating excessive runtime increases. For Random Forest, we explored parameters including tree depth, minimum samples per leaf, minimum samples per split, and feature subsampling. After running some tests and finalizing our choice of Random Forest, we really increased the number of options for tuning in order to give us better results in our final deep dive analysis.

5.2 Overall Results

In our evaluation, we looked at accuracy, precision, recall, F1, and ROC-AUC, with F1 serving as the primary model selection metric. Accuracy was included for completeness, but it was not emphasized because it can be misleading when one class is more common than the other. Precision and recall were both important because this task requires understanding the tradeoff between correctly identifying monetary-relief complaints and avoiding excessive false positives. ROC-AUC was included to assess the model’s ranking ability across thresholds rather than at only a single classification cutoff. Ultimately, we chose F1 as we wanted to balance the precision (correctly classifying monetary relief to better inform consumers) with recall (capturing all monetary relief classifications) because it was important to us to both give consumers good information and simultaneously not discourage them from submitting their complaints based on the prediction results. In **Tables 1** and **2** below, we can see how these metrics compare across model families for holdout and 5-fold cross-validation performance.

Looking at the results, we can see that overall the Random Forest model performed the best with an F1 score of 0.45 and a standard deviation of 0.003 on the 5-fold cross-validated results. Interestingly, the main driver of this performance was the ability of the model to correctly find true positives relative to the other model families, as seen in the higher precision component of 0.37.

Rank	Model	F1 Score	Accuracy	Precision	Recall	ROC-AUC
1	Random Forest	0.4567	0.7816	0.375	0.5839	0.7981
2	Logistic Regression	0.4299	0.6817	0.2992	0.7636	0.7901
3	Support Vector Machine	0.4274	0.6745	0.2954	0.7726	0.7901
4	KNN	0.3268	0.7329	0.2706	0.4124	0.6758

Table 1 – Holdout Performance by F1 Score

Rank	Model	F1 Score	Accuracy	Precision	Recall	ROC-AUC
1	Random Forest	0.4532 ± 0.0037	0.7818 ± 0.0016	0.3739 ± 0.0028	0.5751 ± 0.0071	0.7974 ± 0.0026
2	Logistic Regression	0.4331 ± 0.0044	0.6842 ± 0.0030	0.3017 ± 0.0030	0.7675 ± 0.0102	0.7890 ± 0.0034
3	Support Vector Machine	0.4300 ± 0.0038	0.6761 ± 0.0017	0.2972 ± 0.0023	0.7772 ± 0.0101	0.7891 ± 0.0036

4	KNN	0.3276 ± 0.0146	0.7361 ± 0.0116	0.2737 ± 0.0162	0.4084 ± 0.0110	0.6737 ± 0.0132
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Table 2 – 5-Fold Cross-Validation Performance Mean +/- Standard Deviation

In order to fully understand how all four of the models performed, we added a simple dummy baseline to see if our modeling is actually useful for making predictions. In **Table 3** below, we can see that the stratified dummy classifier only had an F1 score of 0.14 on the holdout F1 and a similarly low score for both precision and recall components. This baseline helped us verify that our models above do indeed add value in our goal to predict monetary response potential for consumers when they fill out a full consumer complaint.

Model	CV F1 (± SD)	Holdout F1	Holdout Accuracy	Holdout Precision	Holdout Recall	Holdout ROC-AUC
Dummy Stratified	0.1584 ± 0.0055	0.1485	0.7309	0.1477	0.1492	0.4943
Dummy Most Frequent	0.0000 ± 0.0000	0	0.8428	0	0	0.5

Table 3 – Dummy Classifier Baseline

5.3 In-Depth Evaluation

5.3.1. Feature Importance & Ablation Analysis

After selecting the Random Forest as our best model, we examined which features contributed most strongly to predictive performance. The feature importance analysis showed that the complaint categories, such as issue, company, and product, were important. Additionally, the simple narrative features also scored high in terms of feature importance. As seen in **Figure 1** below, the top tree-based feature importances are related to the company, issue, product, time, and narrative length features. This suggests that the model is learning some useful distinctions from the complaint and is considering multiple parts of the complaint in order to correctly classify the target.

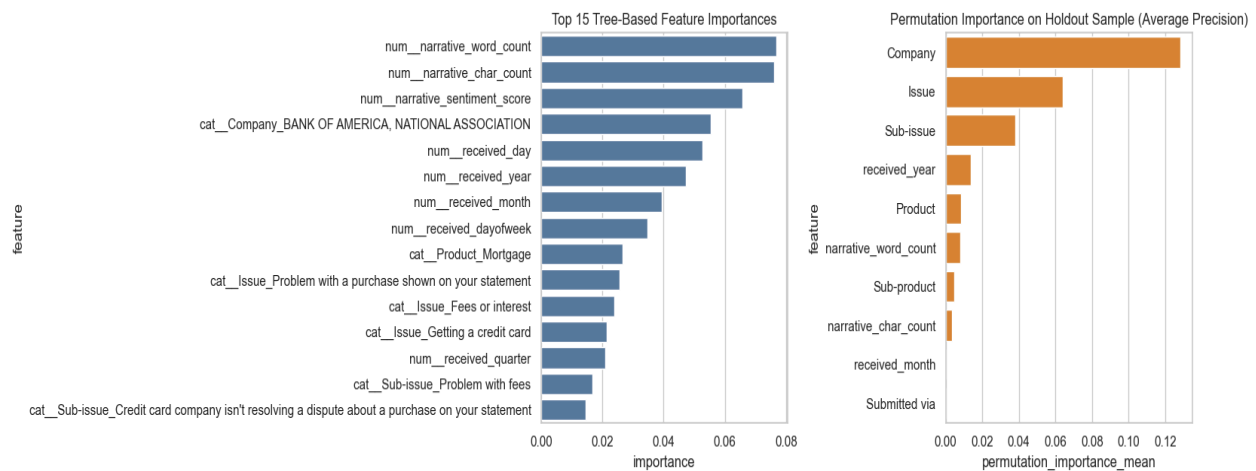


Figure 1 – Feature Importances & Permutation Importance

However, we also performed another scikit-learn inspection for permutation importance to better understand which variables the model actually relied upon when making new predictions on unseen (shuffled) data. Permutation importance works by randomly shuffling one feature at a time and measuring how much the model's performance drops. If performance drops substantially after a feature is shuffled, that feature is contributing a useful signal for our predictive model. If performance does not change much, then the model is not dependent on that variable (Scikit-Learn). This makes permutation importance

especially helpful because it reflects contribution to predictive performance rather than only how often a feature was used in internal tree splits. In our case, the permutation results helped distinguish features that were genuinely driving holdout performance from those that appeared prominent in the tree structure but added less unique signal once the full feature set was considered. As seen in **Figure 1** above, the features that were found to be most useful across shuffles were Company, Issue, and Sub-issue. From which we can reason that these features drive more performance on unseen data rather than the other high-scoring feature importance scores, which express more about the forest's training splits.

In addition to feature and permutation importances, we used ablation analysis by systematically removing groups of features and then re-evaluating model performance. This was useful because feature importance alone does not always show whether a feature group truly matters in combination with the rest of the representation. In this manner, our permutation analysis and ablation analysis seek to remedy the shortcomings of feature importances. The ablation results, seen in **Table 4** below, indicate that some feature groups, including Issue and Company, were meaningfully supporting the F1 performance with roughly a 0.09 and 0.12 drop in F1 score, respectively, when either feature is excluded. On the other hand, features such as Narrative Size, Sentiment, and Product had much less impact on the F1 performance, with only 0.0004, 0.0008, and 0.0026 reductions, respectively.

Rank	Ablation	CV F1	F1 Drop vs Full Model	CV Precision	CV Recall	Features Remaining
1	No Narrative Size	0.457	-0.0038	0.3633	0.6159	13
2	All Features	0.4532	0	0.3739	0.5751	15
3	No Sentiment	0.4531	0.0001	0.3662	0.5942	14
4	No Product Context	0.4508	0.0023	0.3701	0.5767	13
5	No Issue Context	0.3634	0.0898	0.3435	0.3859	13
6	No Company	0.3287	0.1245	0.3433	0.3153	14

Table 4 – Ablation Analysis

5.3.2 Sensitivity Analysis

To understand whether the Random Forest model was robust and not overly dependent on any one hyperparameter choice, we performed a sensitivity analysis by varying some hyperparameter values for the trees, such as the maximum depth and minimum samples per leaf. The goal was to check whether the chosen model configuration was the best or whether slightly different settings performed similarly. As seen below in **Figure 2**, our results show that while some parameter changes reduced performance slightly, the selected configuration was still the strongest among the varying parameter grid. This was reassuring for our model as it suggests that the final model was not performing well only due to a single hyperparameter choice, but instead it benefited from being tuned slightly.

Additionally, the sensitivity analysis helped frame a practical tradeoff between model complexity and performance. Shallower trees or

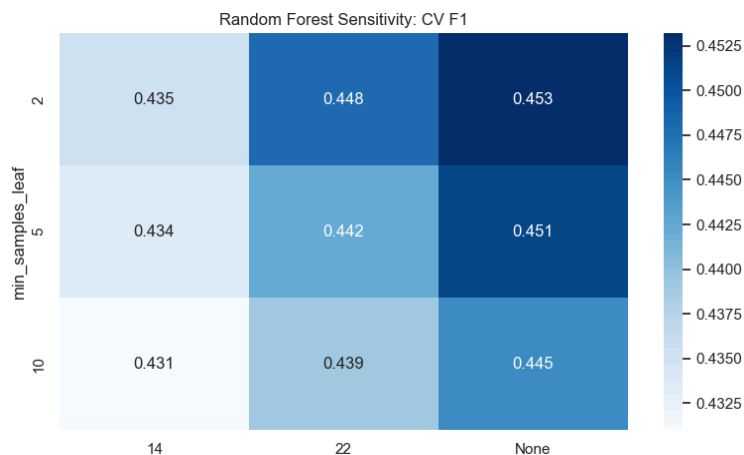


Figure 2 – Sensitivity Analysis

larger leaves reduce overfitting risk and training time, but usually at the cost of performance. Given that our model already took a long time to train, the final selected settings preserved a good F1 score without needing an excessively large or unstable configuration of the random forest.

5.3.3 Tradeoffs

In our final evaluation, we identified several tradeoffs. First, there was a slight tradeoff in precision and recall when changing the classification threshold. The default threshold of 0.50 did not produce the best F1 score. In **Figures 3** and **4** below, we can see that adjusting the decision threshold down slightly to 0.481 marginally improved the balance between precision and recall. In **Figure 3**, we see a rather consistent tradeoff between precision and recall, which resulted in only a slight adjustment of the classification threshold to get a marginally higher F1 score. However, more interestingly, we can see in **Figure 4** a better picture of what happened. With the adjusted classification threshold, we observe roughly 100 more true positives and 100 fewer false negatives, but at the same time, we see roughly 400 more false positives and 400 fewer true negatives. Since we are optimizing for F1 as explained in the beginning, this outcome worked in our favor, as shifting the decision threshold maximized our F1 performance. Therefore, our model was tuned to be more precise in monetary predictions and simultaneously does not discourage consumers from submitting their complaints.

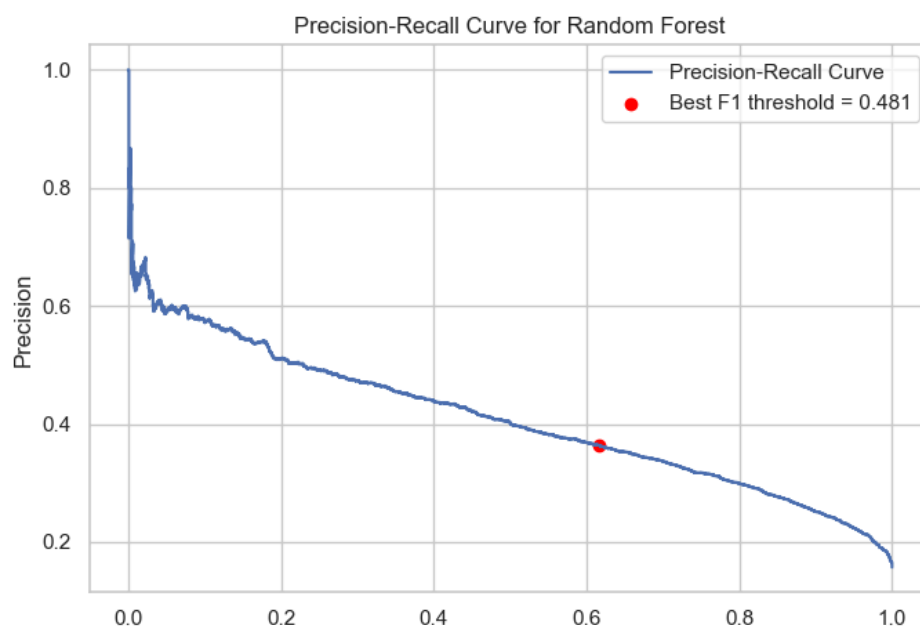


Figure 3 – Precision-Recall Curve

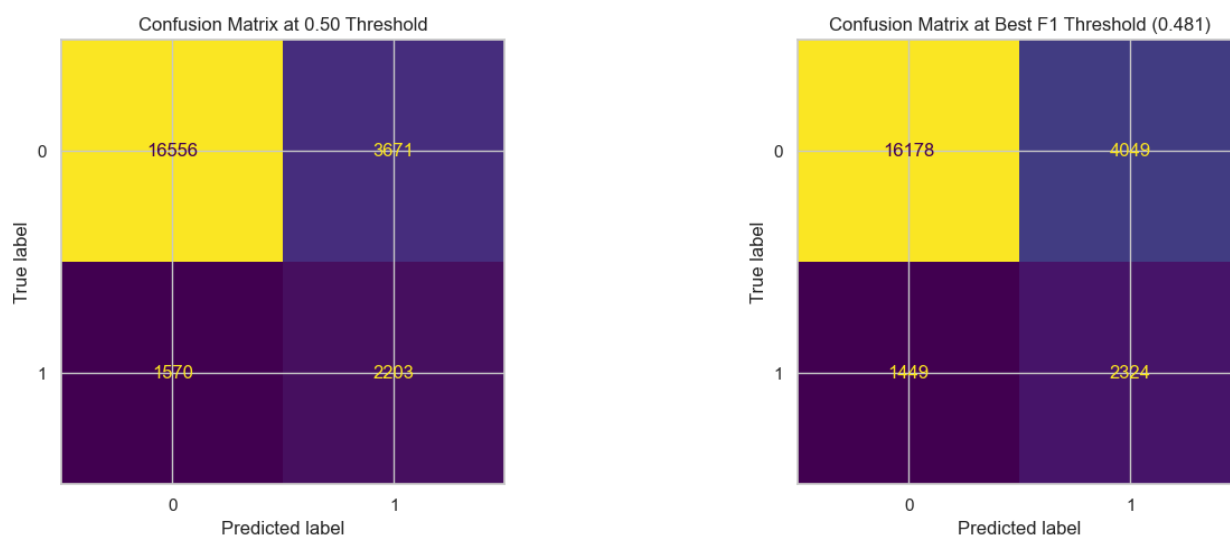


Figure 4 – Confusion Matrix

Second, we tested performance across varying training data sizes. The learning-curve analysis below in **Figure 5** highlights a tradeoff between our training data size and validation performance. While training F1 remained somewhat higher than validation F1 by roughly 0.25, both curves flattened fairly quickly. This suggests that simply adding more of the same type of data may not deliver large gains, and that future improvements are more likely to come from better feature quality or richer representations than from raw data quantity alone. In other words, the model appears to be constrained more by signal quality than by a lack of training examples.

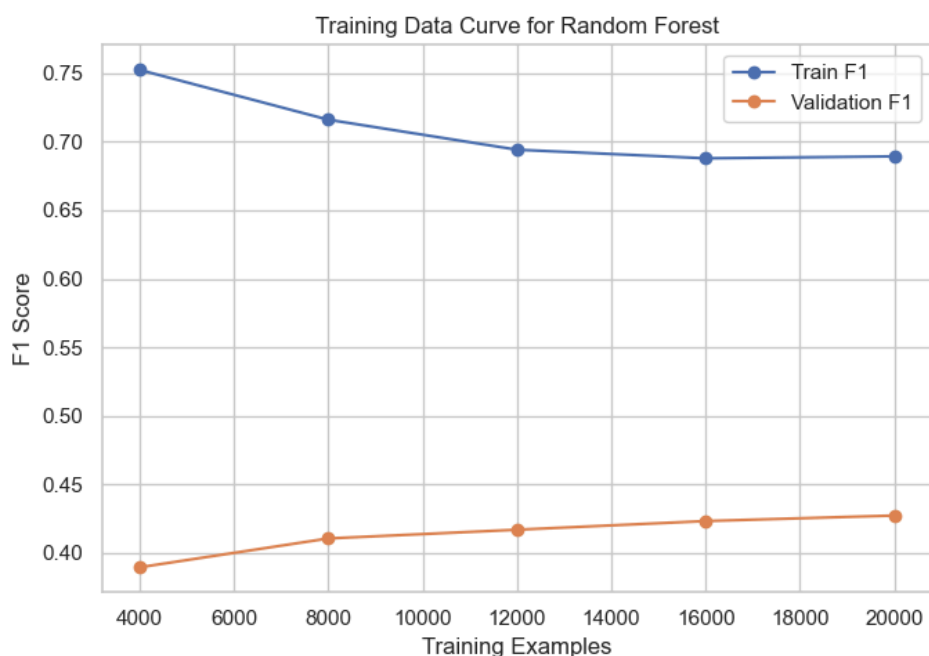


Figure 5 – Training Data Curve

5.4 Failure Analysis

In addition to the model training and performance metrics, we also examined three different specific complaints that the final model made incorrect predictions on. We grouped these failures into three different categories so we could analyze their differences.

First, we encountered a case that we classified as a “likely systemic minority class miss case”. In this case, the complaint was actually for monetary relief but was predicted to be negative. This case suggests that the model may struggle with misunderstanding certain positive-class patterns. In this case specifically, the class prediction probability was 0.499, which shows just how close it came to being correctly allocated to the positive class. A likely improvement for this failure would be finding richer features that better separate the relief-related issues, company-specific behaviors, or alternative narrative unsupervised modeling approaches.

Second, we encountered a case that we classified as an “aggressive positive flag case”. In this case, the model predicted monetary relief for a complaint that was actually closed without relief. This suggests that certain feature patterns that our model used overlap between our two target classes. In this instance, the class prediction probability was 0.871. This classification issue might especially be the case when complaints share similar issues or company feature sets, but have different outcomes. A future fix here might involve better calibration or threshold tuning, or some interaction features that differentiate complaints that otherwise would appear to be similar.

Third, we encountered a case that we classified as a “sparse information edge case”. In this case, the complaint contained less information, and the available features appeared to provide a weak signal. This case is particularly important because the model cannot infer much without good-quality features, particularly related to potentially useful features for the current model, like the narrative component.

Unlike the other two examples above, this example had essentially nothing written into its complaint narrative. For this reason, future improvements for this case and others like it could include gathering better data upstream for better narrative features or utilizing a separate model that handles detailed narratives, which might allow the model to differentiate between instances of feature quality mismatches as seen in the narrative feature examples.

Overall, the failure analysis suggests that the current model performs well when the complaint context is well represented by the structured features in the CFPB dataset and is weakest when the positive class signal is subtle, overlapping, or poorly described. This can inform future work that focuses on richer narrative representations, improved feature interactions, and possibly a multi-model solution that allows for differences in the base feature set and the more detailed narrative sets, particularly when narrative quality differs greatly across training examples.

6. Unsupervised Learning

We had two goals in mind for unsupervised learning: (1) to see if features could be generated from the text data that would help the supervised learning model, and (2) to identify themes in the narratives that would shed light on customer issues. As our first method, we chose K-Means clustering because it can be used to group similar word vectors using distance measures, from which the top terms can be extracted, conceptually similar to a word cloud (Collins-Thompson, 2025). We selected LDA as our second method as it is purpose-built for text modeling and is useful for surfacing themes in a set of documents. LDA is a probabilistic method in which the algorithm estimates the latent topic structure most likely to have generated the observed corpus of text (Collins-Thompson, 2025).

6.1 Applying K-Means Clustering to Consumer Narrative Text

Our workflow began with K-Means clustering. The K-Means clustering method requires numerical features, which for text data is achieved using vectors. We chose to apply the TF-IDF vectorizer here due to its ability to distinguish rare terms relative to the simpler Bag of Words approach. We used the resulting document-term matrix as our final feature representation and trained a series of K-Means models, one for each value of k from 2 to 20 in increments of 2. First, we generated an elbow plot to try to visually find the optimal value. **Figure 6** shows a slight bend at 16 or 18. Given this, we conducted a parameter search over a narrower range (10 to 20, increments of 2) to compare silhouette scores. We chose to use silhouette scores given that it is a preferred evaluation method when ground truth labels are not available. As demonstrated by Collins-Thompson (2025), the silhouette score is a valuable tool for determining the optimal number of clusters when comparing models.

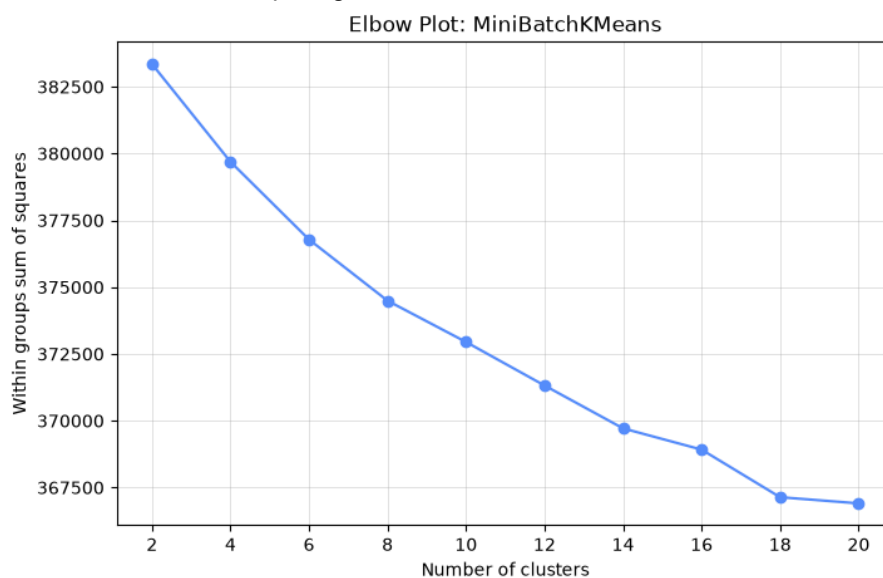


Figure 6 – Elbow Plot for Multiple Values of k

The silhouette score for observation i is defined as: $s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$, where $a(i)$ is the average dissimilarity of instance i with others in the same cluster, and $b(i)$ is the lowest average dissimilarity between the instance and neighboring clusters. These scores range from -1 to 1, where values near -1 indicate weakly clustered elements and values near 1 indicate well-clustered elements. A score near 0 means that the instance is near a decision boundary, where it is nearly as similar to instances in a neighboring cluster as it is to those in its own cluster (Collins-Thompson, 2025). The silhouette scores in our case were quite low. The highest score was for $k = 20$ with a score of 0.025. However, we chose $k = 18$ as our preferred K-Means model given that (1) it was consistent with the elbow plot, (2) its silhouette score was a close second at 0.024, and (3) fewer topics are easier for human comprehension.

Given that many of the clusters had company names (e.g., Chase, Wells Fargo) as top terms, we tried excluding these at the TF-IDF stage to see how performance would change. Applying these stop words caused a decrease in silhouette scores at $k = 18$, from 0.024 to 0.012. However, when visualizing with UMAP, the clusters generated with company stopwords are more informative. **Figure 7** shows the tight interconnection between the clusters, which is in line with the low silhouette scores. If we look at the clusters furthest north and furthest south, we see that the most dissimilar narratives have to do with overdrafts (Cluster 16) and mortgages (Clusters 9 and 14). Although we originally planned to use cluster membership and distance to centroid as features in supervised learning, we ultimately decided to exclude them due to the low silhouette scores.

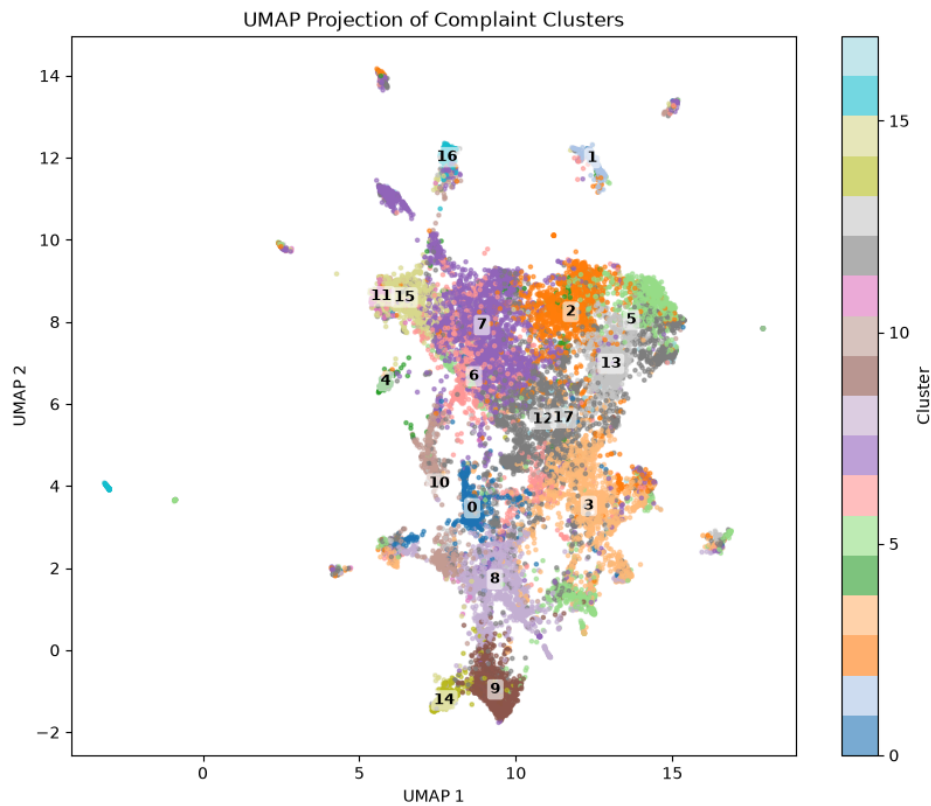


Figure 7 – UMAP Projection of K-Means Clusters (Company Stopwords)

6.2 Utilizing LDA for Topic Discovery in Complaints

The next phase of our workflow was to train the LDA model. As LDA requires a term frequency matrix as input, we used CountVectorizer to create the final feature matrix and applied the company stopwords. However, passing nearly 400,000 narratives into the vectorizer proved to be memory-prohibitive, so we started with a sample of 50,000. We fit a series of LDA models on the sample, searching for the optimal K

topics. We chose C_v topic coherence as our evaluation method since it has the strongest correlation with human coherence ratings across well-known corpora relative to other methods (Röder et al., 2015). Topic coherence takes into account semantic similarities, unlike other metrics such as perplexity. Although there is no pre-defined scale, higher coherence scores are considered better and can be used to optimize the value of K (Collins-Thompson, 2025).

We conducted our parameter search on values of K ranging from 5 to 30 in increments of 5. We selected the highest coherence score (0.51), which was for $K = 20$, and transformed the remaining narratives with this model. **Figure 8** shows virtually all of the topics clustered together via multi-dimensional scaling with the exception of Topic 10, which features top terms like “mortgage”, “loan,” and “escrow.” This suggests that the topics represented in the dataset are quite similar, with the mortgage topic being the most distinct.

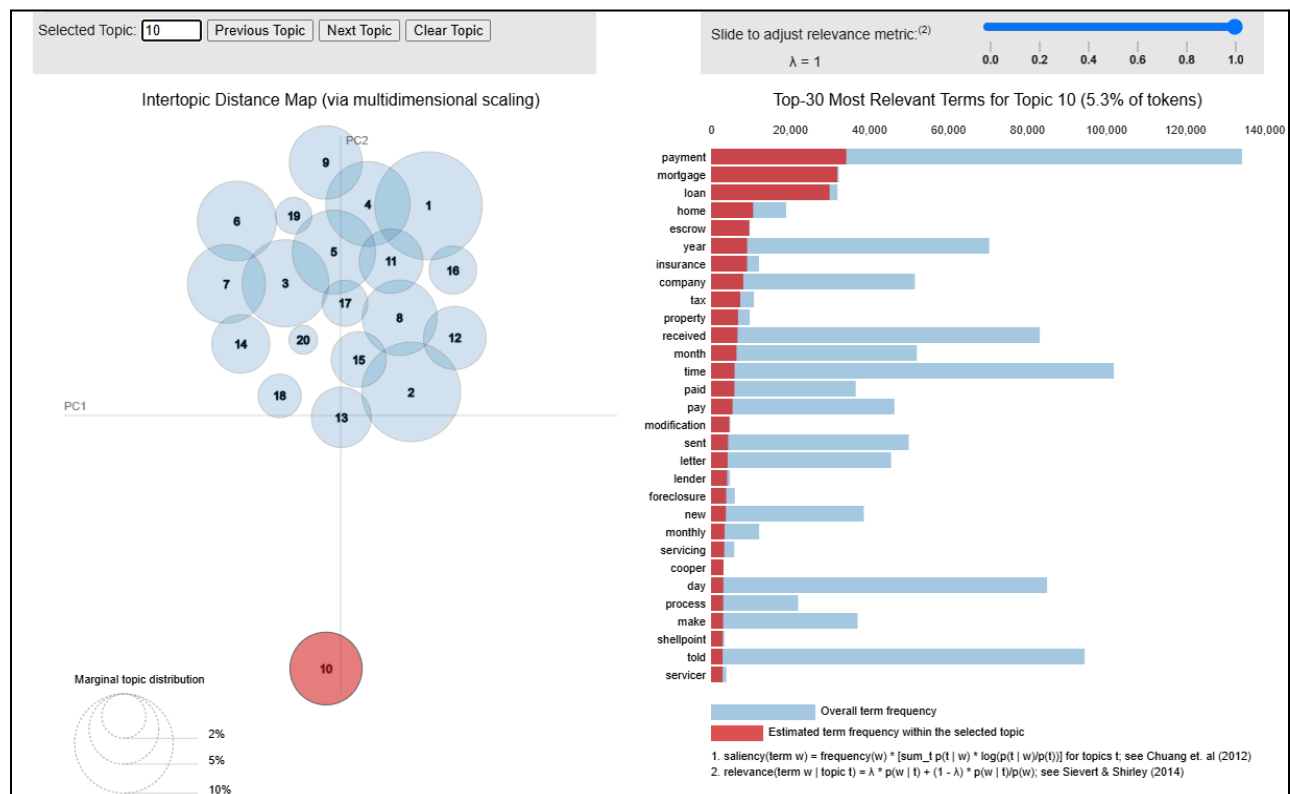


Figure 8 – Intertopic Distance Map and Top Terms generated with pyLDAvis

The value of using LDA on a corpus is that it can surface themes without requiring a human to manually read every document. In the case of the CFPB complaint database, there are nearly 400,000 unique narratives in our timeframe, so LDA can add significant value. At the outset of our review, we only knew that the corpus included complaints about products in the five areas we selected: mortgage, checking or savings accounts, credit cards, prepaid cards, and bank accounts. Out of the 20 topics generated by our LDA model, 50% of them were informative beyond the initial list of product names (see **Appendix C** for the full list of topics and terms). For example, Topic 7 involves merchant disputes, and Topic 17 pertains to bonus offers. Other notable topics are Topic 16, which pertains to ATMs and cash withdrawals at branch locations, and Topic 19, which is about travel-related refunds. The most interesting topic uncovered was Topic 14 (“debt,” “consumer,” “law”), which upon inspection of sample narratives, revealed consumers alleging that their banks were threatening them with legal consequences due to a perceived breach.

For additional insight into the complaint topics, we used the relevance parameter (λ) within pyLDAvis, which determines the weight given to the probability of a topic term relative to its lift (Sievert & Shirley,

2014). As we lower λ towards zero, we will see terms that are more distinctive within the topic. At $\lambda = 1.0$, the top three terms for Topic 9 are "time", "like," and "money." After changing the view to $\lambda = 0.2$, we find terms that suggest family financial issues, such as "mother," "father," "death," and "elderly." Shifting our focus to Topic 1, the initial top terms are not very informative: "told," "said," and "called." But if we set $\lambda = 0$, terms like "nasty," "irritated," and "mad" become more apparent. Taken together, this suggests the presence of complaints about contentious phone calls. Indeed, a quick random sample of three narratives assigned to this topic reveals consumers dissatisfied with customer service representatives who treated them "horrible" or called at unreasonable hours.

Finally, for each narrative, we added the dominant topic and the probabilities for all 20 topics to the dataframe for consideration as features in our supervised learning models.

6.3 Model Comparison and Sensitivity Analysis

Overall, it is difficult to perform a direct performance comparison between the two types of unsupervised models as their evaluation metrics are fundamentally different. The silhouette scores used for K-Means are constrained to a scale of -1 to 1, while topic coherence has no defined scale in the academic literature. Informal community guidance suggests that C_v values around 0.50 indicate moderate performance, close to our best LDA model's score of 0.51 (Sara, 2019). In contrast, our best K-Means model ($k = 18$) resulted in a low silhouette score (0.024), suggesting weak separation between clusters. Taken together, these results suggest that the LDA model is more effective for identifying themes in the consumer narratives.

Figure 9 shows that the LDA model was not very sensitive to changes during the parameter search. The best model with $K = 20$ had a coherence score of 0.51 and perplexity of 860.5 (lower is better). Increasing K to 30 only dropped the coherence by 0.04 and decreased perplexity by 3.7, while decreasing K to 10 dropped the coherence by 0.01 and increased perplexity by 33.3.

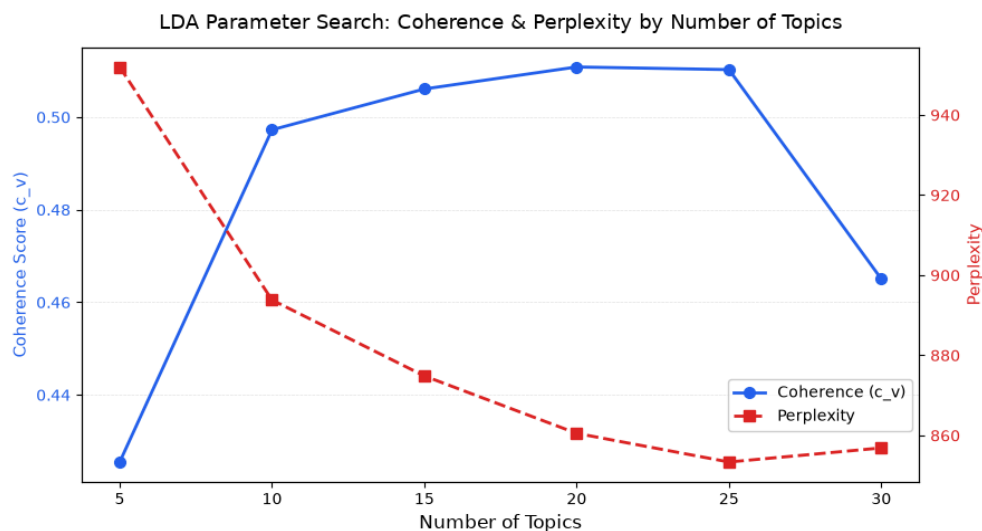


Figure 9 – LDA Scoring Metrics by Number of Topics (K)

7. Discussion

7.1 Supervised Learning

From the supervised learning portion of our project, we learned that predicting whether a CFPB banking complaint ends in monetary relief is possible, but it is not an easy classification problem. Our best performing models were Random Forest classifiers, which gave the strongest overall F1 performance on the holdout data. In the main final model, the Random Forest reached about 0.45 F1 in its best

configuration, while the narrative-informed version reached about 0.46 F1 in its best configuration. That told us the signal is real, but also that the outcome is only partially predictable from the structured complaint fields and our lightweight narrative features.

What surprised us most was that the narrative features helped less than we had originally expected. We thought sentiment and narrative length would produce a larger gain, but the ablation results showed that removing sentiment or narrative-size features barely changed F1. In contrast, removing Company, Issue, or Sub-issue caused the biggest performance drops. That suggests the complaint context and institution involved were more predictive than simple text-derived signals or even our more advanced LDA topics.

The biggest challenges were class imbalance, missing or inconsistent narrative coverage, and the tradeoff between recall and precision. Because monetary relief is relatively rare, it was easy for a model to look good on accuracy while failing at the actual task. We responded by using stronger evaluation metrics, comparing against dummy baselines, and testing multiple model families instead of assuming one approach would work. We also built both an all-records model and a narrative-present subset model (which we used for our final model) so we could separate the effect of missing text from the effect of the narrative features themselves. When it came to model results, we ran into a consistent tradeoff between precision and recall. Thankfully, it fit well with our F1 focus, but it was still something we had to wrestle with as there was no easy way to improve either component without sacrificing the other.

With more time and resources, we would extend the solution in three main ways. First, we would use richer text modeling, such as transformer embeddings or maybe a more finance-tuned language model, instead of only sentiment, length, clustering, and topic features. Second, we would do more temporal and out-of-sample validation to test how well the model generalizes across years and changing complaint patterns, since we didn't look much into training on only recent data. Third, we would spend more time on calibration and decision-threshold tuning so the model could be better tailored to a real use case, such as prioritizing high recall or high precision, but for our general case, the balanced F1 score worked well. Overall, supervised learning showed us that the problem is learnable, but the biggest gains will likely come from better text representations and more careful deployment that addresses end users' current needs.

7.2 Unsupervised Learning

Applying K-Means clustering and LDA to the text of the consumer narratives suggested relative uniformity in the topic space, which is not surprising given that we are working with a corpus that exclusively contains financial product complaints with very similar terminology. This likely explains the weak separation of clusters in the K-Means model (silhouette score: 0.024). Visualizing the data with UMAP and pyLDAviz seemed to confirm this, showing substantial overlap of clusters and topics, with the exception of mortgage. The topics generated by LDA were more meaningful than K-Means and allowed us to better understand some of the consumers' frustrations in the complaints. The top terms in the LDA topics suggested issues with merchants, ATMs, and travel-related refunds. By sampling a handful of narratives and filtering to distinctive terms with pyLDAviz, we were able to uncover sub-topics such as family financial issues and legal threats allegedly made to consumers by financial institutions. The LDA model's coherence score of 0.51 was consistent with moderate performance.

Given the low performance for K-Means, we did not include derived features such as cluster ID and distance to centroid in the supervised model. We did, however, include dominant topic and topic probabilities from the LDA model as features, hypothesizing that there would be some signal in the narratives that could differentiate the types of requests that tended to secure monetary relief. We were surprised to see the mean F1 of the best supervised model only increased by about 0.008, from 0.4567 to 0.4649 on the holdout dataset.

The primary challenges we encountered were memory limitations in training the models given our narrative population of 398,004 records after de-duplication and dropping nulls. For K-Means, our workaround was to utilize MiniBatchKMeans with a batch size of 1,000. For LDA, our first roadblock was passing the full population through CountVectorizer. We resolved this by vectorizing a sample of 50,000

narratives, then fitting the LDA model on those. After the model was fit, we applied the topic model to the remaining records in the dataframe in order to assign topic probabilities to the full set of narratives. Parameter searches were also costly to conduct, so we made sure to write results to disk to avoid repeated runs.

If we had additional time and resources, we would have conducted more trial-and-error with different stopwords and expanded the number of topics to focus on hidden themes, rather than prioritizing the highest scores. Bastani et al. (2019) increased their topics to 40 but judged their model on meaningfulness to humans, rather than coherence scores or perplexity. In our case, we were also attempting to generate features for supervised learning, so we had to balance searching for meaningful themes with quantitative measures that we hoped would translate to better results downstream.

8. Ethical Considerations

As stated, our goal is to help consumers better understand the complaint process and generate a prediction as to what type of response they will receive from the company - monetary or non-monetary. This is intended to encourage consumers to speak up, make their voices heard, and hold financial institutions accountable when abuse is present. With our model, a false positive would be predicting monetary relief for a consumer when none actually happens. A false negative would be predicting no monetary relief when relief is actually given. In order to balance these concerns, we prioritized the F1 score when selecting our best supervised model.

Our unsupervised models leveraged text data that is already scrubbed of personal information by the CFPB, so individual privacy is protected. Although the CFPB requires consumers to provide consent for their narratives to be made public, we recognize that consumers did not contemplate their narratives being passed into machine learning algorithms for deeper analysis. In the event that we ever made our predictions public-facing, we would mitigate this by offering consumers the ability to opt out of being included in the model.

Statement of Work

- **Noah:** Data extraction, data exploration, supervised learning - feature engineering, supervised model development and evaluation, preparation of GitHub repo, supervised content in written report
- **Tony:** Data exploration, unsupervised learning - feature engineering, unsupervised model development and evaluation, unsupervised content, editing, and formatting of written report

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Project Submission

GitHub repository: https://github.com/n-tamm/cfpb_mortgage_complaints

APPENDIX A - Final Feature List

```
structured_numeric = [  
    "received_year",  
    "received_month",  
    "received_quarter",  
    "received_dayofweek",  
    "received_day",  
    "narrative_present",  
    "narrative_char_count",  
    "narrative_word_count",  
    "narrative_sentiment_score",  
]
```

```
rf_categorical = [  
    "Product",  
    "Sub-product",  
    "Issue",  
    "Sub-issue",  
    "Company",  
    "Submitted via",  
]
```

APPENDIX B - Supervised Model Results with LDA Features

Rank	Model	F1 Score	Accuracy	Precision	Recall	ROC-AUC
1	Random Forest	0.4649	0.7782	0.3745	0.613	0.8053
2	Logistic Regression	0.4461	0.6945	0.312	0.7824	0.805
3	Support Vector Machine	0.4438	0.689	0.3087	0.789	0.8049
4	KNN	0.391	0.7315	0.3038	0.5481	0.7398

Table 1B – Holdout Performance by F1 Score

Rank	Model	F1 Score	Accuracy	Precision	Recall	ROC-AUC
1	Random Forest	0.4611 ± 0.0030	0.7796 ± 0.0012	0.3745 ± 0.0023	0.5999 ± 0.0051	0.8026 ± 0.0014
2	Logistic Regression	0.4441 ± 0.0023	0.6948 ± 0.0028	0.3111 ± 0.0020	0.7758 ± 0.0054	0.8031 ± 0.0023
3	Support Vector Machine	0.4430 ± 0.0021	0.6898 ± 0.0024	0.3086 ± 0.0018	0.7850 ± 0.0052	0.8030 ± 0.0026
4	KNN	0.3894 ± 0.0158	0.7305 ± 0.0054	0.3024 ± 0.0100	0.5471 ± 0.0320	0.7362 ± 0.0119

Table 2B – 5-Fold Cross-Validation Performance Mean +/- Standard Deviation

APPENDIX C - LDA Top Terms by Topic

Topic <i>pyLDAviz</i> Numbering	top_terms
1	told, said, called, money, time, day, asked, got
2	account, check, fund, money, checking, deposit, closed, day
3	payment, credit, card, fee, account, late, pay, time
4	account, number, card, email, phone, address, received, information
5	account, financial, issue, matter, request, regarding, complaint, despite
6	account, fraud, claim, fraudulent, transaction, report, card, credit
7	dispute, merchant, charge, refund, credit, received, claim, provided
8	card, charge, credit, purchase, transaction, debit, called, time
9	time, like, money, help, need, work, year, people
10	payment, mortgage, loan, home, escrow, year, insurance, company
11	complaint, cfpb, practice, despite, failure, violation, written, request
12	fee, point, customer, service, term, reward, representative, offer
13	credit, report, reporting, account, information, consumer, application, bureau
14	debt, consumer, law, security, account, state, right, notice
15	balance, statement, payment, synchrony, charge, month, paid, purchase
16	cash, atm, branch, money, deposit, app, withdrawal, transaction
17	account, bonus, offer, pnc, day, deposit, opened, promotion
18	document, sent, copy, letter, form, application, submitted, requested
19	refund, ticket, flight, hotel, travel, card, credit, trip
20	chime, paypal, car, rental, company, bofa, repair, vehicle